

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-I (ALL STAR BATCH)]
ANSWER KEY
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	B	A	C	B	A	C	B	A
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	D	D	A	A	A	B	C	A	C

PART-2 : CHEMISTRY

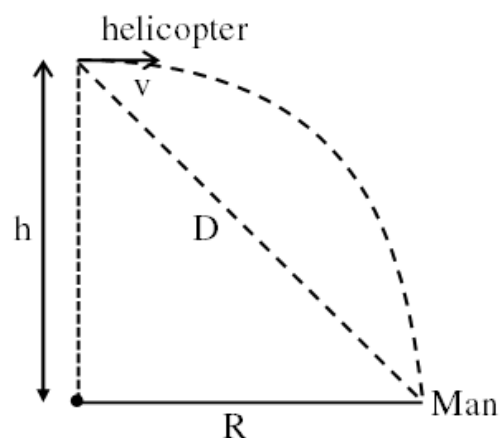
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	D	B	B	A	C	B	D	B	C
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	C	A	B	C	A	D	D	C	A

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	B	A	D	C	C	B	B	C
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	D	D	C	A	B	C	B	C	B

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

1. Ans (C)



$$R = \sqrt{\frac{2h}{g}} \cdot v$$

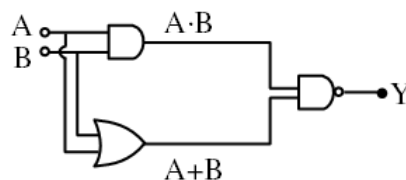
$$D = \sqrt{R^2 + h^2}$$

$$= \sqrt{\left(\sqrt{\frac{2h}{g}} \cdot v\right)^2 + h^2}$$

$$D = \sqrt{\frac{2hv^2}{g} + h^2}$$

Option (3) is correct

2. Ans (C)



$$Y = (A \cdot B) \cdot (A + B)$$

$$Y_{(0,0)} = 1$$

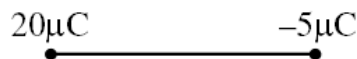
$$Y_{(0,1)} = 1$$

$$Y_{(1,0)} = 1$$

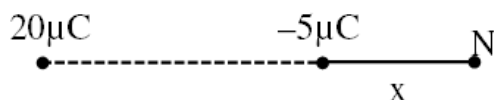
$$Y_{(1,1)} = 0$$

Option (3) is correct

3. Ans (B)



Null point is possible only right side of $-5\mu\text{C}$



$$E_N = +\frac{k(-5\mu\text{C})}{x^2} + \frac{k(20\mu\text{C})}{(5+x)^2} = 0$$

$$x = 5 \text{ cm}$$

$\therefore$ option (2) is correct

4. Ans (A)

T_2 = sink temperature

$$\eta = 1 - \frac{T_2}{T_1}$$

$$\frac{1}{4} = 1 - \frac{T_2}{T_1}$$

$$\frac{T_2}{T_1} = \frac{3}{4} \dots (i)$$

$$\frac{1}{2} = 1 - \frac{T_2 - 58}{T_1}$$

$$\frac{T_2}{T_1} - \frac{58}{T_1} = \frac{1}{2}$$

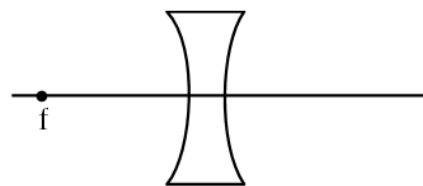
$$\frac{3}{4} = \frac{58}{T_1} + \frac{1}{2}$$

$$\frac{1}{4} = \frac{58}{T_1} \Rightarrow T_1 = 232$$

$$T_2 = \frac{3}{4} \times 232$$

$$T_2 = 174 \text{ K}$$

5. Ans (C)



$$U = -f$$

$$\frac{1}{V} - \frac{1}{U} = \frac{1}{-f} \Rightarrow \frac{1}{V} = -\frac{2}{f}$$

$$V = \frac{-f}{2}$$

$$m = \frac{V}{U} = \frac{1}{2}$$

$$\text{distance} = \frac{f}{2}$$

Option (3)

6. Ans (B)

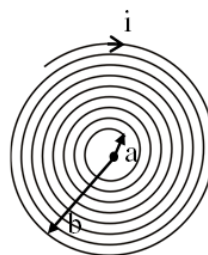


Initially no. of atoms of B = 0 after $t = 0$, no. of atoms of B will start increasing & reaches maximum value when rate of decay of B = rate of formation of B.

After that maximum value, no. of atoms will start decreasing as growth & decay both are exponential functions, so best possible graph is (2)

Option (2)

7. Ans (A)



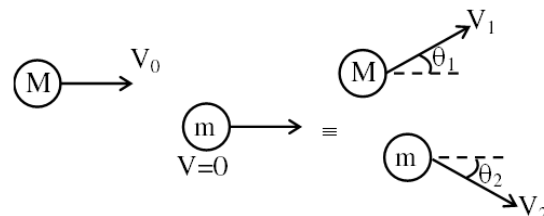
$$\text{No. of turns in } dx \text{ width} = \frac{N}{b-a} dx$$

$$\int dB = \int_a^b \left(\frac{N}{b-a} \right) dx \frac{\mu_0 i}{2x}$$

$$B = \frac{N\mu_0 i}{2(b-a)} \ln \left(\frac{b}{a} \right)$$

Option (1)

8. Ans (C)



$$\text{given } \theta_1 = \theta_2 = \theta$$

from momentum conservation

$$\text{in x-direction } MV_0 = MV_1 \cos \theta + mV_2 \cos \theta$$

$$\text{in y-direction } 0 = MV_1 \sin \theta - mV_2 \sin \theta$$

Solving above equations

$$V_2 = \frac{MV_1}{m}, V_0 = 2V_1 \cos \theta$$

From energy conservation

$$\frac{1}{2} MV_0^2 = \frac{1}{2} MV_1^2 + \frac{1}{2} mV_2^2$$

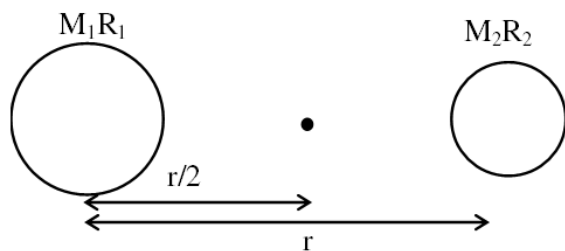
Substituting value of V_2 & V_0 , we will get

$$\frac{M}{m} + 1 = 4\cos^2 \theta \leq 4$$

$$\frac{M}{m} \leq 3$$

Option (3)

9. Ans (B)



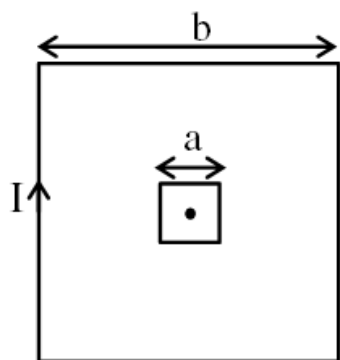
$$\frac{1}{2}mV^2 - \frac{GM_1m}{r/2} - \frac{GM_2m}{r/2} = 0$$

$$\frac{1}{2}mV^2 = \frac{2Gm}{r}(M_1 + M_2)$$

$$V = \sqrt{\frac{4G(M_1 + M_2)}{r}}$$

Option (2)

10. Ans (A)



$$B = \left[\frac{\mu_0}{4\pi} \frac{I}{b/2} \times 2 \sin 45^\circ \right] \times 4$$

$$\phi = 2\sqrt{2} \frac{\mu_0}{\pi} \frac{I}{b} \times a^2$$

$$\therefore M = \frac{\phi}{I} = \frac{2\sqrt{2}\mu_0 a^2}{\pi b} = \frac{\mu_0}{4\pi} 8\sqrt{2} \frac{a^2}{b}$$

Option (1)

11. Ans (A)

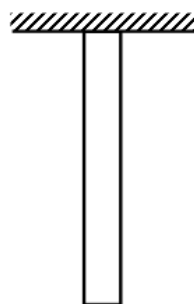
When $V_i > 3$ volt, $V_R > 0$

Because diode will be in forward biased state

When $V_i \leq 3$ volt; $V_R = 0$

Because diode will be in reverse biased state.

12. Ans (D)



We know,

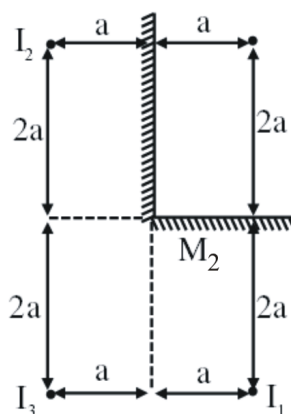
$$\Delta l = \frac{WL}{2AY}$$

$$\Delta l = \frac{10 \times 1}{2 \times 5} \times 100 \times 10^{-4} \times 2 \times 10^{11}$$

$$\Delta l = \frac{1}{2} \times 10^{-9} = 5 \times 10^{-10} \text{ m}$$

Option (4)

13. Ans (D)



Shortest distance is $2a$ between I_1 & I_3

But answer given is for I_1 & I_2

$$\sqrt{(4a)^2 + (2a)^2}$$

$$a\sqrt{20}$$

$$4.47 a$$

Option (2)

14. Ans (A)

torque $\tau \rightarrow ML^2T^{-2}$ (III)

Impulse $I \Rightarrow MLT^{-1}$ (I)

Tension force $\Rightarrow MLT^{-2}$ (IV)

Surface tension $\Rightarrow MT^{-2}$ (II)

Option (1)

15. Ans (A)

$$\int_{p_0}^p \frac{dp}{p} = -a \int_0^v dv$$

$$\ln \left(\frac{p}{p_0} \right) = -av$$

$$p = p_0 e^{-av}$$

For temperature maximum p-v product should be maximum

$$T = \frac{pv}{nR} = \frac{p_0 v e^{-av}}{R}$$

$$\frac{dT}{dv} = 0 \Rightarrow \frac{p_0}{R} \{ e^{-av} + v e^{-av} (-a) \}$$

$$\frac{p_0 e^{-av}}{R} \{ 1 - av \} = 0$$

$$v = \frac{1}{a}, \infty$$

$$T = \frac{p_0}{Rae} = \frac{p_0}{Rae}$$

at $v = \infty$

$$T = 0$$

Option (1)

16. Ans (A)

$$(i) \frac{\pi p a^4}{8 \eta L} = \frac{dv}{dt} = \text{Volumetric flow rate}$$

(poiseuille's law)

$$(ii) h \rho g = \frac{2s}{r} \cos \theta$$

$$(iii) \text{RHS} \Rightarrow \epsilon \times \frac{1}{4\pi\epsilon_0} \frac{a}{r^2} \times \frac{1}{\epsilon} = \frac{q}{t} \times \frac{1}{r^2}$$

$$= \frac{I}{L^2} = IL^{-2}$$

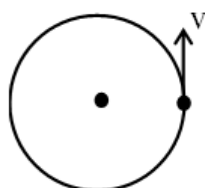
LHS

$$T = \frac{I}{A} = IL^{-2}$$

$$(iv) W = \tau \theta$$

Option (1)

17. Ans (B)



$$|\vec{L}| = mvr$$

And direction will be upward & remain constant

Option (2)

18. Ans (C)

$$Z = \sqrt{(X_L - X_C)^2 + R^2} = R \because X_L = X_C$$

Option (3)

19. Ans (A)

$$\lambda_P = \frac{h}{P_P} \quad \lambda_e = \frac{h}{P_e}$$

$$\therefore \lambda_P = \lambda_e$$

$$\Rightarrow P_P = P_e$$

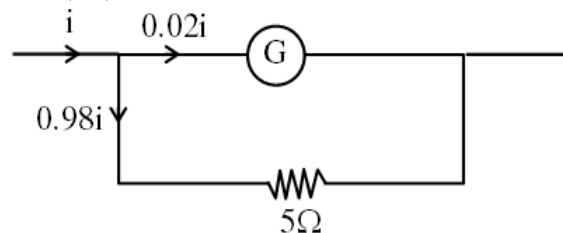
$$(K)_P = \frac{P_P^2}{2m_P}$$

$$(K)_e = \frac{P_e^2}{2m_e}$$

$$K_P < K_e \text{ as } m_P > m_e$$

Option (1)

20. Ans (C)



$$0.02i R_g = 0.98i \times 5$$

$$R_g = 245 \Omega$$

Option (3)

PART-1 : PHYSICS

SECTION-II

1. Ans (500)

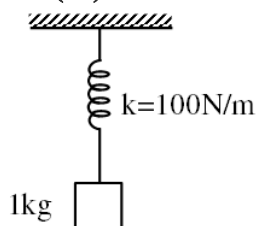
$$B = - \frac{\Delta P}{\left(\frac{\Delta v}{v} \right)} = - \frac{\rho gh}{\left(\frac{\Delta v}{v} \right)}$$

$$- \frac{B \frac{\Delta v}{v}}{\rho g} = h$$

$$\frac{9.8 \times 10^8 \times 0.5}{100 \times 10^3 \times 9.8} = h$$

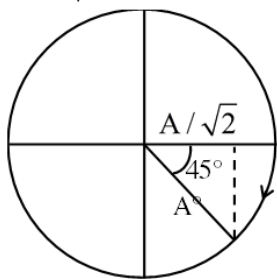
$$h = 500$$

2. Ans (8)



$$KE = PE$$

$$y = \frac{A}{\sqrt{2}} = A \sin \omega t$$



$$t = \frac{T}{8} = \frac{T}{x}$$

$$x = 8$$

3. Ans (64)

$$h_T = h_R = 160 \dots (i)$$

$$d = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

$$d = \sqrt{2R} [\sqrt{h_T} + \sqrt{h_R}]$$

$$d = \sqrt{2R} [\sqrt{x} + \sqrt{160-x}]$$

$$\frac{d(d)}{dx} = 0$$

$$\frac{1}{2\sqrt{x}} + \frac{1(-1)}{2\sqrt{160-x}} = 0$$

$$\frac{1}{\sqrt{x}} = \frac{1}{\sqrt{160-x}}$$

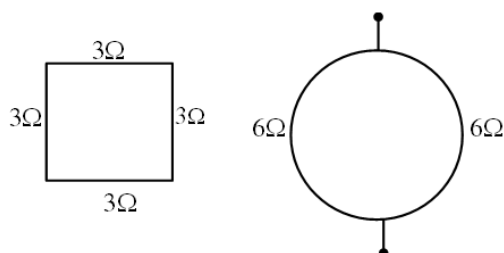
$$x = 80 \text{ m}$$

$$d_{\max} = \sqrt{2 \times 6400} \left[\sqrt{\frac{80}{1000}} + \sqrt{\frac{20}{1000}} \right]$$

$$= \frac{80\sqrt{2} \times 2\sqrt{80}}{10\sqrt{10}}$$

$$= 8 \times 2 \times \sqrt{2} \times 2\sqrt{2} = 64 \text{ km}$$

4. Ans (3)



$$R_{eq} = 3\Omega$$

5. Ans (10)

$$\mu = 9.0 \times 10^{-4} \frac{\text{kg}}{\text{m}}$$

$$T = 900 \text{ N}$$

$$V = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{900}{9 \times 10^{-4}}} = 1000 \text{ m/s}$$

$$f_1 = 500 \text{ Hz}$$

$$f = 550$$

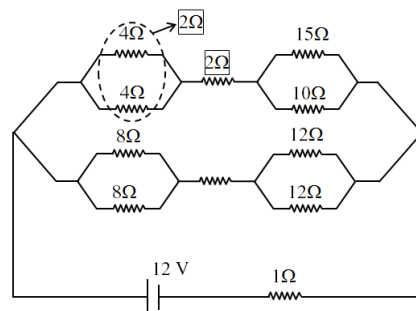
$$\frac{nV}{2\ell} = 500 \dots (i)$$

$$\frac{(n+1)V}{2\ell} = 500 \dots (ii)$$

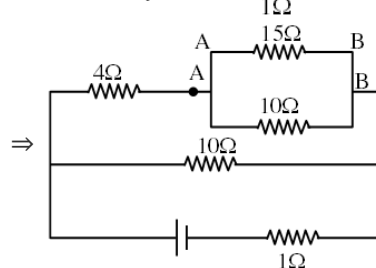
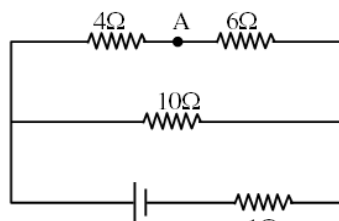
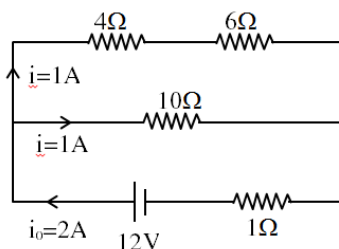
$$(ii) - (i) \quad \frac{V}{2\ell} = 50$$

$$\ell = \frac{1000}{2 \times 50} = 10$$

6. Ans (6)



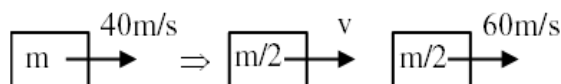
⇒ effective circuit diagram will be



$$\text{Point drop across } 6\Omega = 1 \times 6 = 6 = V_{AB}$$

$$\Rightarrow \text{Hence point drop across } 15\Omega = 6 \text{ volt} = V_{AB}$$

7. Ans (1)



$$P_i = P_f$$

$$m \times 40 = \frac{m}{2} \times v + \frac{m}{2} \times 60$$

$$40 = \frac{v}{2} + 30$$

$$\Rightarrow v = 20$$

$$(K.E.)_i = \frac{1}{2} m \times (40)^2 = 800m$$

$$(K.E.)_f = \frac{1}{2} \frac{m}{2} \cdot (20)^2 + \frac{1}{2} \cdot \frac{m}{2} (60)^2 = 1000m$$

$$|\Delta K.E.| = |1000m - 800m| = 200m$$

$$\frac{\Delta K.E.}{(K.E.)_i} = \frac{200m}{800m} = \frac{1}{4} = \frac{x}{4}$$

$$x = 1$$

8. Ans (500)

$$E = 50 \sin\left(\omega t - \frac{\omega}{c} \cdot x\right)$$

$$\text{Energy density} = \frac{1}{2} \epsilon_0 E_0^2$$

$$\text{Energy for volume } V = \frac{1}{2} \epsilon_0 E_0^2 \cdot V = 5.5 \times 10^{-12}$$

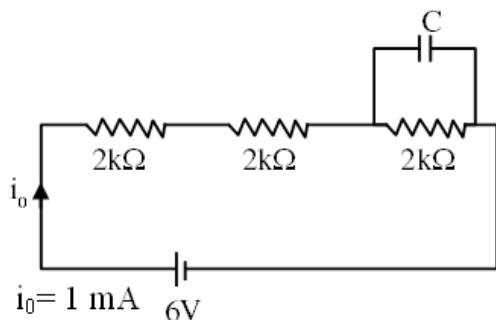
$$\frac{1}{2} 8.8 \times 10^{-12} \times 2500V = 5.5 \times 10^{-12}$$

$$V = \frac{5.5 \times 2}{2500 \times 8.8} = .0005m^3$$

$$= .0005 \times 10^6 (\text{c.m})^3$$

$$= 500 (\text{c.m})^3$$

9. Ans (100)



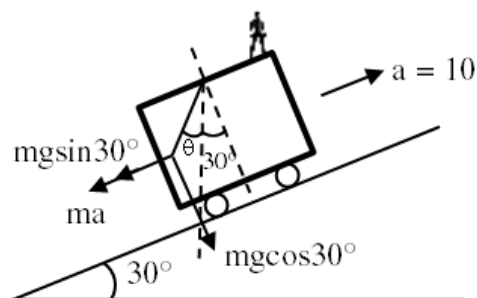
$$i_0 = 1 \text{ mA} \quad 6V$$

$$\text{Pot. Diff. across each resistor} = 2V$$

$$q = CV$$

$$= 50 \times 10^{-6} \times 2 = 100 \times 10^{-6} = 100 \mu C$$

10. Ans (30)



$$\tan(30 + \theta) = \frac{mg \sin 30^\circ + ma}{mg \cos 30^\circ}$$

$$\tan(30 + \theta) = \frac{5 + 10}{5\sqrt{3}} = \frac{1 + 2}{\sqrt{3}}$$

$$\frac{\tan \theta + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}} \tan \theta} = \sqrt{3}$$

$$\sqrt{3} \tan \theta + 1 = 3 - \sqrt{3} \tan \theta$$

$$2\sqrt{3} \tan \theta = 2$$

$$\tan \theta = \frac{1}{\sqrt{3}}$$

$$\theta = 30^\circ$$

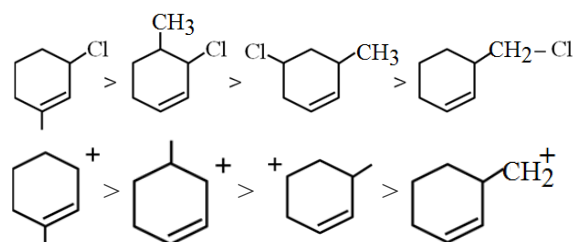
PART-2 : CHEMISTRY

SECTION-I

1. Ans (A)

As it is example of SN^1 .

so carbocation stability $\uparrow$, reaction rate $\uparrow$

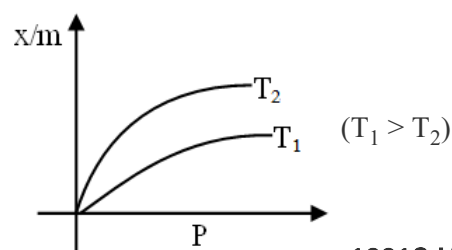


2. Ans (D)

$$\frac{x}{m} \propto P^{1/n} \quad \left(0 < \frac{1}{n} < 1\right)$$

On Increasing temperature $\frac{x}{m}$ decreases.

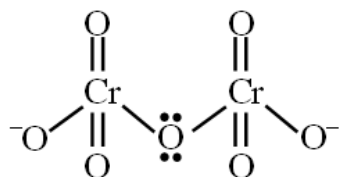
$\therefore$ adsorption is generally exothermic



3. **Ans (B)**

Major component of portland cement is "Tricalcium silicate (51% , $3\text{CaO} \cdot \text{SiO}_2$)

4. **Ans (B)**



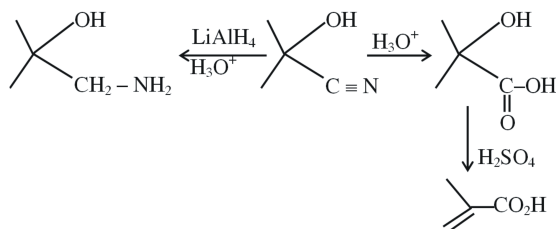
dichromate ion contain non-linear symmetrical Cr–O–Cr Bond

5. **Ans (A)**

In Lactose it is β $\text{C}_1 - \text{C}_4$ glycosidic linkage.

In Maltose, Amylose α $\text{C}_1 - \text{C}_4$ glycosidic linkage is present

6. **Ans (C)**



7. **Ans (B)**

Ytterbium shows +2 oxidation state with diamagnetic nature.

So ans is 2

8. **Ans (D)**

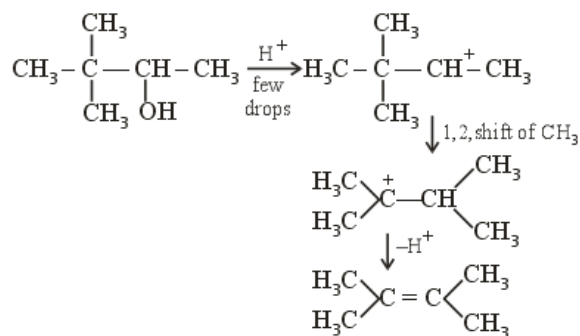
(A) Aluminium is reactive metal so Aluminium is extracted by electrolysis of Alumina with molten mixture of Cryolite

(B) Cryolite, Na_3AlF_6

Here Al is in +3 O.S.

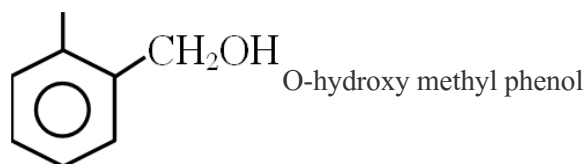
So Answer is 4

9. **Ans (B)**



10. **Ans (C)**

Monomer of Novolac is

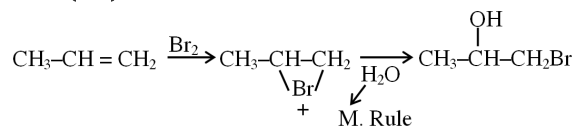


11. **Ans (B)**

The process of producing syn-gas from coal is called gasification of coal.

Syn-gas having composition of CO & H_2 in 1 : 1

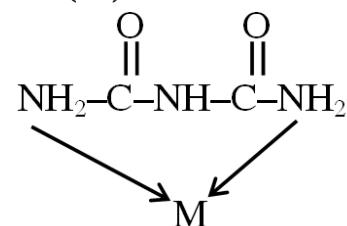
12. **Ans (C)**



Its IUPAC name 1-bromopropan-2-ol

A and R are true and (R) is the correct explanation of (A)

13. **Ans (A)**



Biuret :- Bidentate ligand

The denticity of organic ligand is 2

14. **Ans (B)**

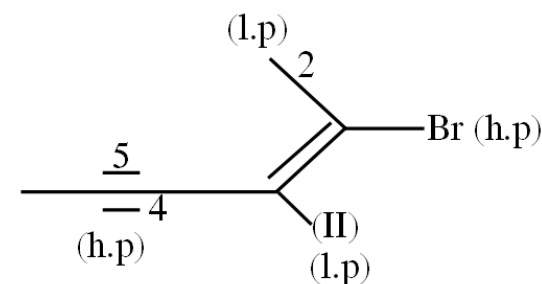
From left to right in periodic table :-

Metallic character decreases

Non-metallic character increases

$\Rightarrow$ It is due to increase in ionization enthalpy and increase in electron gain enthalpy.

15. **Ans (C)**



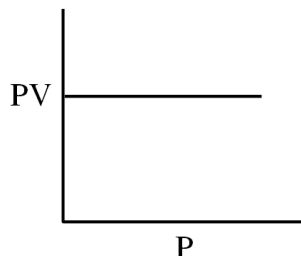
h.p. $\Rightarrow$ higher priority
l.p. $\Rightarrow$ lower priority

2E -2- bromo hex -2- en-4-yne

16. **Ans (A)**

$$PV = nRT \text{ (n,T constant)}$$

$$PV = \text{constant}$$



17. **Ans (D)**

Both assertion & reason are correct & (R) is the correct explanation of (A)

18. **Ans (D)**

∴ Van't Hoff factor is highest for KHSO_4

∴ colligative property (ΔT_f) will be highest for

KHSO_4

19. **Ans (C)**

BOD values of clean water (A) is less than 5 ppm

So $A < 5$

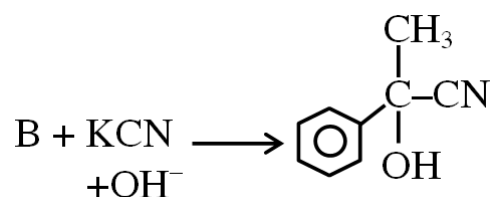
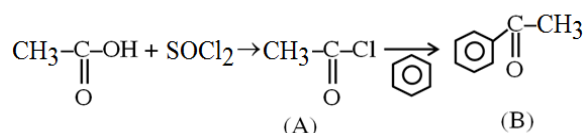
BOD values of polluted water (B) is greater than

17 ppm

So $B > 17$

So Ans. is 3

20. **Ans (A)**



PART-2 : CHEMISTRY

SECTION-II

1. **Ans (25)**

$$\Delta G^\circ = -nFE^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$\begin{aligned} &= \frac{\Delta H^\circ + nFE^\circ}{T} \\ &= \frac{(-825.2 \times 10^3) + (2 \times 96487 \times 4.315)}{298} \\ &= \frac{-825.2 \times 10^3 + 832.682 \times 10^3}{298} \\ &= \frac{7.483 \times 10^3}{298} = 25.11 \text{ JK}^{-1}\text{mol}^{-1} \end{aligned}$$

∴ Nearest integer answer is 25

2. **Ans (20)**

$$[\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}] = \frac{\text{weight}/M_w}{V \text{ (L)}}$$

$$\Rightarrow x \times 10^{-2} = \frac{6.3/126}{250/1000}$$

$$x = 20$$

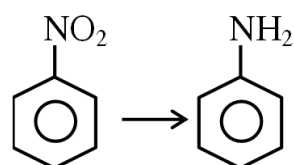
3. **Ans (4)**

Pbs, CuS , As_2S_3 , CdS are soluble in 50%

HNO_3 HgS , Sb_2S_3 are insoluble in 50% HNO_3

So Answer is 4.

4. **Ans (5)**



Reagents used can be

(i) $\text{Sn} + \text{HCl}$

(ii) $\text{Fe} + \text{HCl}$

(iii) $\text{Zn} + \text{HCl}$

(iv) $\text{H}_2 - \text{Pd}$

(v) H_2 (Raney Ni)

5. **Ans (3)**

The number of halogen forming halic (V) acid

HClO_3

HBrO_3

HIO_3

So Answer is 3

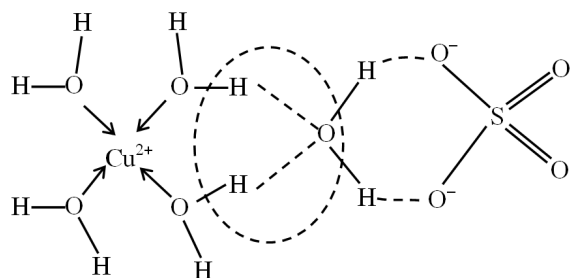
6. Ans (2)

$$k = \frac{2.303}{t} \log \frac{a}{a-x}$$

$$\frac{2.303}{t_{50\%}} \log \frac{100}{100-50} = \frac{2.303}{t_{75\%}} \log \frac{100}{100-75}$$

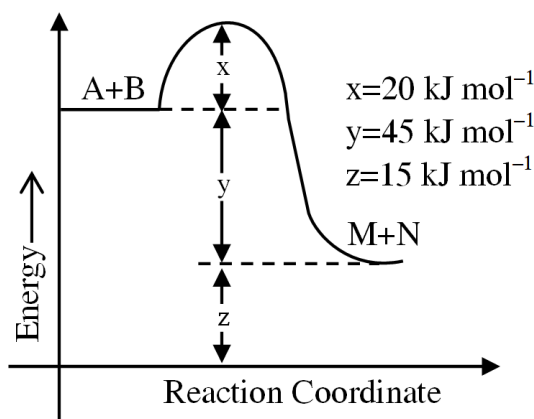
$$t_{75\%} = 2 t_{50\%}$$

7. Ans (1)



One hydrogen bonded H_2O molecule

8. Ans (45)



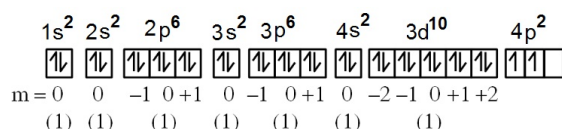
$$\Delta H = E_{a_f} - E_{a_b}$$

$$= 20 - 65$$

$$= -45 \text{ KJ/mol}$$

$$|\Delta H| = 45 \text{ KJ/mol}$$

9. Ans (7)



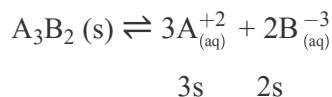
Completely filled orbital with $m_l = 0$ are

$$= 1 + 1 + 1 + 1 + 1 + 1 + 1$$

$$= 7$$

So Answer is 7

10. Ans (108)



$$K_{SP} = (3s)^3 (2s)^2$$

$$K_{SP} = 108 s^5 \text{ \& } s = (X/M)$$

$$K_{SP} = 108 \left(\frac{x}{m} \right)^5$$

$$\text{given } K_{SP} = a \left(\frac{x}{m} \right)^5$$

comparing $a = 108$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (C)

$(p \wedge \sim q) \rightarrow (p \vee q)$ is tautology

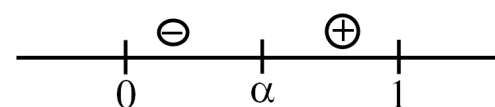
p	q	$\sim q$	$p \wedge \sim q$	$p \vee q$	$(p \wedge \sim q) \rightarrow (p \vee q)$
T	T	F	F	T	T
T	F	T	T	T	T
F	T	F	F	T	T
F	F	T	F	F	T

2. Ans (C)

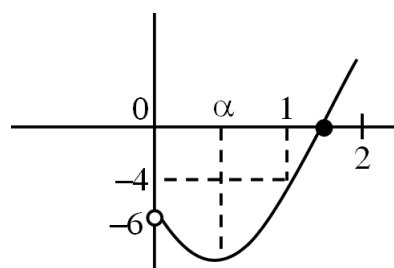
Let $e^x = t > 0$

$$f(t) = t^4 + 2t^3 - t - 6 = 0$$

$$f'(t) = 4t^3 + 6t^2 - 1$$



$$f''(t) = 12t^2 + 12t > 0$$



$$f(0) = -6, f(1) = -4, f(2) = 24$$

$\Rightarrow$ Number of real roots = 1

3. **Ans (B)**

$$S = \frac{2^2 - 1^2}{1^2 \times 2^2} + \frac{3^2 - 2^2}{2^2 \times 3^2} + \frac{4^2 - 3^2}{3^2 \times 4^2} + \dots$$

$$= \left[\frac{1}{1^2} - \frac{1}{2^2} \right] + \left[\frac{1}{2^2} - \frac{1}{3^2} \right] + \left[\frac{1}{3^2} - \frac{1}{4^2} \right] + \dots + \left[\frac{1}{10^2} - \frac{1}{11^2} \right]$$

$$= 1 - \frac{1}{121}$$

$$= \frac{120}{121}$$

4. **Ans (A)**

Equation of plane is

$$3x - 2y + 4z - 7 + \lambda(x + 5y - 2z + 9) = 0$$

$$(3 + \lambda)x + (5\lambda - 2)y + (4 - 2\lambda)z + 9\lambda - 7 = 0$$

passing through (1, 4, -3)

$$\Rightarrow 3 + \lambda + 20\lambda - 8 - 12 + 6\lambda + 9\lambda - 7 = 0$$

$$\Rightarrow \lambda = \frac{2}{3}$$

$\Rightarrow$ equation of plane is

$$-11x - 4y - 8z + 3 = 0$$

$$\Rightarrow \alpha + \beta + \gamma = -23$$

5. **Ans (D)**

$$\int_0^x \sqrt{1 - (f'(t))^2} dt = \int_0^x f(t) dt \quad 0 \leq x \leq 1$$

differentiating both the sides

$$\sqrt{1 - (f'(x))^2} = f(x)$$

$$\Rightarrow 1 - (f'(x))^2 = f^2(x)$$

$$\frac{f'(x)}{\sqrt{1 - f^2(x)}} = 1$$

$$\sin^{-1} f(x) = x + C$$

$$\because f(0) = 0 \Rightarrow C = 0 \Rightarrow f(x) = \sin x$$

$$\text{Now } \lim_{x \rightarrow 0} \frac{\int_0^x \sin t dt}{x^2} \left(\frac{0}{0} \right) = \frac{1}{2}$$

6. **Ans (C)**

$$|3\vec{a} + \vec{b}|^2 = |2\vec{a} + 3\vec{b}|^2$$

$$(3\vec{a} + \vec{b}) \cdot (3\vec{a} + \vec{b}) = (2\vec{a} + 3\vec{b}) \cdot (2\vec{a} + 3\vec{b})$$

$$9\vec{a} \cdot \vec{a} + 6\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{b} = 4\vec{a} \cdot \vec{a} + 12\vec{a} \cdot \vec{b} + 9\vec{b} \cdot \vec{b}$$

$$5|\vec{a}|^2 - 6\vec{a} \cdot \vec{b} = 8|\vec{b}|^2$$

$$5(8)^2 - 6.8|\vec{b}| \cos 60^\circ = 8|\vec{b}|^2 \quad \left(\because \frac{1}{8}|\vec{a}| = 1 \right)$$

$$\Rightarrow |\vec{a}| = 8$$

$$40 - 3|\vec{b}| = |\vec{b}|^2$$

$$\Rightarrow |\vec{b}|^2 + 3|\vec{b}| - 40 = 0$$

$$|\vec{b}| = -8, \quad |\vec{b}| = 5$$

(rejected)

7. **Ans (C)**

$$f(x) = |(x-3)(x+1)| \cdot e^{(3x-2)^2}$$

$$f(x) = \begin{cases} (x-3)(x+1) \cdot e^{(3x-2)^2} & ; x \in (3, \infty) \\ -(x-3)(x+1) \cdot e^{(3x-2)^2} & ; x \in [-1, 3] \\ (x-3) \cdot (x+1) \cdot e^{(3x-2)^2} & ; x \in (-\infty, -1) \end{cases}$$

Clearly, non-differentiable at $x = -1$ & $x = 3$.

8. **Ans (B)**

Let numbers be $\frac{a}{r}, a, ar \rightarrow G.P$

$$\frac{a}{r}, 2a, ar \rightarrow A.P \Rightarrow 4a = \frac{a}{r} + ar \Rightarrow r + \frac{1}{r} = 4$$

$$r = 2 \pm \sqrt{3}$$

$$4^{\text{th}} \text{ form of G.P} = 3r^2 \Rightarrow ar^2 = 3r^2 \Rightarrow a = 3$$

$$r = 2 + \sqrt{3}, a = 3, d = 2a - \frac{a}{r} = 3\sqrt{3}$$

$$r^2 - d = (2 + \sqrt{3})^2 - 3\sqrt{3}$$

$$= 7 + 4\sqrt{3} - 3\sqrt{3}$$

$$= 7 + \sqrt{3}$$

9. **Ans (B)**

Note that (1,2) and (2,3) satisfy $0 < |x - y| \leq 1$

but (1,3) does not satisfy it so

$0 \leq |x - y| \leq 1$ is symmetric but not transitive

So, (2) is correct.

10. **Ans (C)**

$$\int \frac{dx}{(x-1)^{3/4}(x+2)^{5/4}}$$

$$= \int \frac{dx}{\left(\frac{x+2}{x-1}\right)^{5/4} \cdot (x-1)^2}$$

$$\text{put } \frac{x+2}{x-1} = t$$

$$= -\frac{1}{3} \int \frac{dt}{t^{5/4}}$$

$$= \frac{4}{3} \cdot \frac{1}{t^{1/4}} + C$$

$$= \frac{4}{3} \left(\frac{x-1}{x+2} \right)^{1/4} + C$$

11. **Ans (A)**

$$\text{First line is } \frac{x}{\sin \alpha} - \frac{y}{\cos \alpha} = \frac{k \cos 2\alpha}{\sin 2\alpha}$$

$$\Rightarrow x \cos \alpha - y \sin \alpha = \frac{k}{2} \cos 2\alpha$$

$$\Rightarrow p = \left| \frac{k}{2} \cos \alpha \right| \Rightarrow 2p = |k \cos 2\alpha| \quad \dots(i)$$

$$\text{second line is } x \sin \alpha + y \cos \alpha = k \sin 2\alpha$$

$$\Rightarrow q = |k \sin 2\alpha| \quad \dots(ii)$$

$$\text{Hence } 4p^2 + q^2 = k^2 \quad (\text{From (i) \& (ii)})$$

12. **Ans (D)**

$$\operatorname{cosec} 18^\circ = \frac{1}{\sin 18^\circ} = \frac{4}{\sqrt{5}-1} = \sqrt{5} + 1$$

$$\text{Let } \operatorname{cosec} 18^\circ = x = \sqrt{5} + 1$$

$$\Rightarrow x - 1 = \sqrt{5}$$

Squaring both sides, we get

$$x^2 - 2x + 1 = 5$$

$$\Rightarrow x^2 - 2x - 4 = 0$$

13. **Ans (D)**

Here

$$D = \begin{vmatrix} 2 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & a \end{vmatrix} = 2(-a-1) - 1(a-1) + 1+1$$

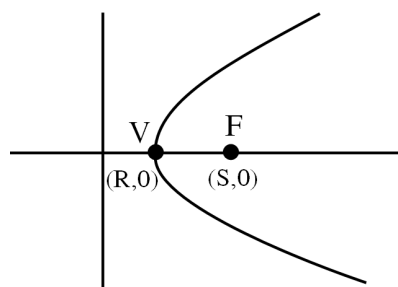
$$= 1-3a$$

$$D_3 = \begin{vmatrix} 2 & 1 & 5 \\ 1 & -1 & 3 \\ 1 & 1 & b \end{vmatrix} = 2(-b-3) - 1(b-3) + 5(1+1)$$

$$= 7-3b$$

for $a = \frac{1}{3}$, $b \neq \frac{7}{3}$, system has no solutions

14. **Ans (C)**



V $\rightarrow$ Vertex

F $\rightarrow$ focus

$$VF = S - R$$

$$\text{So latus rectum} = 4(S - R)$$

15. **Ans (A)**

If $f(x)$ is continuous at $x = 0$, $\text{RHL} = \text{LHL} = f(0)$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{\cos^2 x - \sin^2 x - 1}{\sqrt{x^2+1} - 1} \cdot \frac{\sqrt{x^2+1} + 1}{\sqrt{x^2+1} + 1}$$

(Rationalisation)

$$\lim_{x \rightarrow 0^+} -\frac{2\sin^2 x}{x^2} \cdot (\sqrt{x^2+1} + 1) = -4$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{1}{x} \ln \left(\frac{1 + \frac{x}{a}}{1 - \frac{x}{b}} \right)$$

$$\lim_{x \rightarrow 0^-} \frac{\ln \left(1 + \frac{x}{a} \right)}{\left(\frac{x}{a} \right) \cdot a} + \frac{\ln \left(1 - \frac{x}{b} \right)}{\left(-\frac{x}{b} \right) \cdot b}$$

$$= \frac{1}{a} + \frac{1}{b}$$

$$\text{So } \frac{1}{a} + \frac{1}{b} = -4 = k$$

$$\Rightarrow \frac{1}{a} + \frac{1}{b} + \frac{4}{k} = -4 - 1 = -5$$

16. Ans (B)

$$\frac{dy}{dx} = \frac{2^x 2^y - 2^x}{2^y}$$

$$2^y \frac{dy}{dx} = 2^x (2^y - 1)$$

$$\int \frac{2^y}{2^y - 1} dy = \int 2^x dx$$

$$\frac{\ln(2^y - 1)}{\ln 2} = \frac{2^x}{\ln 2} + C$$

$$\Rightarrow \log_2(2^y - 1) = 2^x \log_2 e + C$$

$$\because y(0) = 1 \Rightarrow 0 = \log_2 e + C$$

$$C = -\log_2 e$$

$$\Rightarrow \log_2(2^y - 1) = (2^x - 1) \log_2 e$$

$$\text{put } x = 1, \log_2(2^y - 1) = \log_2 e$$

$$2^y = e + 1$$

$$y = \log_2(e + 1) \text{ Ans.}$$

17. Ans (C)

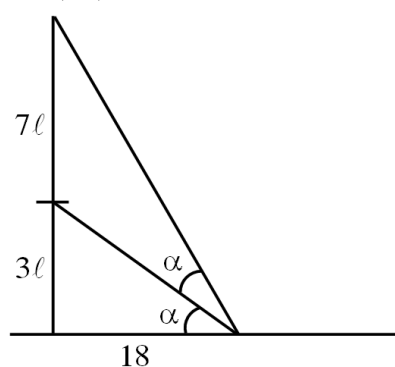
$$\lim_{x \rightarrow 0} \frac{\sin^2(\pi \cos^4 x)}{x^4}$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos(2\pi \cos^4 x)}{2x^4}$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos(2\pi - 2\pi \cos^4 x)}{[2\pi(1 - \cos^4 x)]^2} 4\pi^2 \cdot \frac{\sin^4 x}{2x^4} (1 + \cos^2 x)^2$$

$$= \frac{1}{2} \cdot 4\pi^2 \cdot \frac{1}{2} (2)^2 = 4\pi^2$$

18. Ans (B)



Let height of pole = 10ℓ

$$\tan \alpha = \frac{3\ell}{18} = \frac{\ell}{6}$$

$$\tan 2\alpha = \frac{10\ell}{18}$$

$$\frac{2 \tan \alpha}{1 - \tan^2 \alpha} = \frac{10\ell}{18}$$

$$\text{use } \tan \alpha = \frac{\ell}{6} \Rightarrow \ell = \sqrt{\frac{72}{5}}$$

$$\text{height of pole} = 10\ell = 12\sqrt{10}$$

19. Ans (C)

$a_r = e^{\frac{i2\pi r}{9}}$, $r = 1, 2, 3, \dots$ $a_1, a_2, a_3, \dots$ are in G.P.

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ a_n & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix} = \begin{vmatrix} a_1 & a_2^2 & a_1^3 \\ a_1^4 & a_1^5 & a_1^6 \\ a_1^7 & a_1^8 & a_1^9 \end{vmatrix}$$

$$= a_1 \cdot a_1^4 \cdot a_1^7 \begin{vmatrix} 1 & a_1 & a_1^2 \\ 1 & a_1 & a_1^2 \\ 1 & a_1 & a_1^2 \end{vmatrix} = 0$$

$$\text{Now } a_1 a_9 - a_3 a_7 = a_1^{10} - a_1^{10} = 0$$

20. Ans (B)

$$12x \cos \theta + 5y \sin \theta = 60$$

$$\frac{x \cos \theta}{5} + \frac{y \sin \theta}{12} = 1$$

$$\text{is tangent to } \frac{x^2}{25} + \frac{y^2}{144} = 1$$

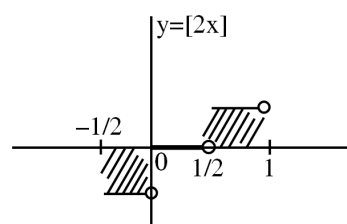
$$144x^2 + 25y^2 = 3600$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (5)

$$I = \int_{-1/2}^1 ([2x] + |x|) dx$$



$$= \int_{-1/2}^0 [2x] dx + \int_0^1 |x| dx$$

$$= 0 + \int_{-1/2}^0 (-x) dx + \int_0^1 x dx$$

$$= \left(-\frac{x^2}{2} \right)_{-1/2}^0 + \left(\frac{x^2}{2} \right)_0^1$$

$$= \left(0 + \frac{1}{8} \right) + \frac{1}{2}$$

$$= \frac{5}{8}$$

$$8I = 5$$

2. **Ans (98)**

Let $z = x + iy$

$$\arg\left(\frac{x-2+iy}{x+2+iy}\right) = \frac{\pi}{4}$$

$$\arg(x-2+iy) - \arg(x+2+iy) = \frac{\pi}{4}$$

$$\tan^{-1}\left(\frac{y}{x-2}\right) - \tan^{-1}\left(\frac{y}{x+2}\right) = \frac{\pi}{4}$$

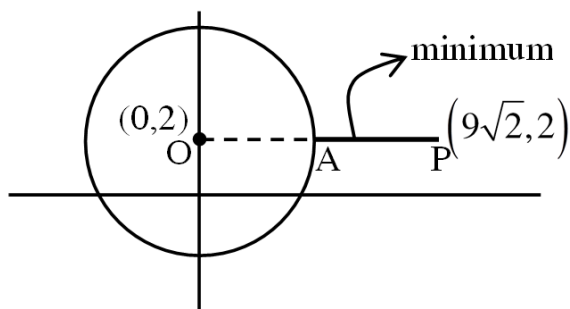
$$\frac{\frac{y}{x-2} - \frac{y}{x+2}}{1 + \left(\frac{y}{x-2}\right) \cdot \left(\frac{y}{x+2}\right)} = \tan \frac{\pi}{4} = 1$$

$$\frac{xy + 2y - xy + 2y}{x^2 - 4 + y^2} = 1$$

$$4y = x^2 - 4 + y^2$$

$$x^2 + y^2 - 4y - 4 = 0$$

locus is a circle with center $(0, 2)$ & radius $= 2\sqrt{2}$



$$\text{min. value} = (AP)^2 = (OP - OA)^2$$

$$= (9\sqrt{2} - 2\sqrt{2})^2$$

$$= (7\sqrt{2})^2 = 98$$

3. **Ans (61)**

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+1}{6} = \lambda$$

$$x = 2\lambda + 1, y = 3\lambda + 2, z = 6\lambda - 1$$

for point of intersection of line & plane

$$2(2\lambda + 1) - (3\lambda + 2) + (6\lambda - 1) = 6$$

$$7\lambda = 7 \Rightarrow \lambda = 1$$

point : $(3, 5, 5)$

$$(\text{distance})^2 = (3+1)^2 + (5+1)^2 + (5-2)^2$$

$$= 16 + 36 + 9 = 61$$

4. **Ans (2)**

$$f(x) = x^2 + ax + 1$$

$$f'(x) = 2x + a$$

when $f(x)$ is increasing on $[1, 2]$

$$2x + a \geq 0 \quad \forall x \in [1, 2]$$

$$a \geq -2x \quad \forall x \in [1, 2]$$

$$R = -4$$

when $f(x)$ is decreasing on $[1, 2]$

$$2x + a \leq 0 \quad \forall x \in [1, 2]$$

$$a \leq -2 \quad \forall x \in [1, 2]$$

$$S = -2$$

$$|R - S| = |-4 + 2| = 2$$

5. **Ans (398)**

$$7 \times 8, 10 \times 10, 13 \times 12, 16 \times 14 \dots\dots$$

$$T_n = (3n + 4)(2n + 6) = 2(3n + 4)(n + 3)$$

$$= 2(3n^2 + 13n + 12) = 6n^2 + 26n + 24$$

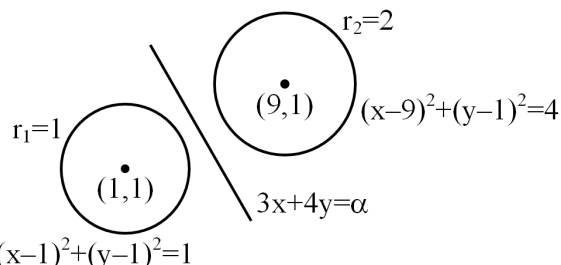
$$S_{10} = \sum_{n=1}^{10} T_n = 6 \sum_{n=1}^{10} n^2 + 26 \sum_{n=1}^{10} n + 24 \sum_{n=1}^{10} 1$$

$$= \frac{6(10 \times 11 \times 21)}{6} + 26 \times \frac{10 \times 11}{2} + 24 \times 10$$

$$= 10 \times 11(21 + 13) + 240 = 3980$$

$$\text{Mean} = \frac{S_{10}}{10} = \frac{3980}{10} = 398$$

6. **Ans (165)**



Both centres should lie on either side of the line as well as line can be tangent to circle.

$$(3 + 4 - \alpha) \cdot (27 + 4 - \alpha) < 0$$

$$(7 - \alpha) \cdot (31 - \alpha) < 0 \Rightarrow \alpha \in (7, 31) \dots(1)$$

d_1 = distance of $(1, 1)$ from line

d_2 = distance of $(9, 1)$ from line

$$d_1 \geq r_1 \Rightarrow \frac{|7 - \alpha|}{5} \geq 1$$

$$\Rightarrow \alpha \in (-\infty, 2] \cup [12, \infty) \dots(2)$$

$$d_2 \geq r_2 \Rightarrow \frac{|31 - \alpha|}{5} \geq 2$$

$$\Rightarrow \alpha \in (-\infty, 21] \cup [41, \infty) \dots(3)$$

$$(1) \cap (2) \cap (3) \Rightarrow \alpha \in [12, 21]$$

$$\text{Sum of integers} = 165$$

7. **Ans (576)**

VOWELS $\rightarrow$ 2 Vowels
 $\rightarrow$ 4 Consonants

All Consonants should not be together
 = Total – All consonants together,
 = $6! - 3! 4! = 576$

8. **Ans (4)**

$$x\phi(x) = \int_5^x 3t^2 - 2\phi'(t) dt$$

$$x\phi(x) = x^3 - 125 - 2[\phi(x) - \phi(5)]$$

$$x\phi(x) = x^3 - 125 - 2\phi(x) + 2\phi(5)$$

$$\phi(0) = 4 \Rightarrow \phi(5) = -\frac{133}{2}$$

$$\phi(x) = \frac{x^3 + 8}{x + 2}$$

$$\phi(2) = 4$$

9. **Ans (55)**

$$\left(\frac{x}{4} - \frac{12}{x^2} \right)^{12}$$

$$T_{r+1} = (-1)^r \cdot {}^{12}C_r \left(\frac{x}{4} \right)^{12-r} \left(\frac{12}{x^2} \right)^r$$

$$T_{r+1} = (-1)^r \cdot {}^{12}C_r \left(\frac{1}{4} \right)^{12-r} (12)^r \cdot (x)^{12-3r}$$

$$\text{Term independent of } x \Rightarrow 12 - 3r = 0 \Rightarrow r = 4$$

$$T_5 = (-1)^4 \cdot {}^{12}C_4 \left(\frac{1}{4} \right)^8 (12)^4 = \frac{3^6}{4^4} \cdot k$$

$$\Rightarrow k = 55$$

10. **Ans (28)**

I_1 = first unit is functioning

I_2 = second unit is functioning

$$P(I_1) = 0.9, P(I_2) = 0.8$$

$$P(\bar{I}_1) = 0.1, P(\bar{I}_2) = 0.2$$

$$P = \frac{0.8 \times 0.1}{0.1 \times 0.2 + 0.9 \times 0.2 + 0.1 \times 0.8} = \frac{8}{28}$$

$$98P = \frac{8}{28} \times 98 = 28$$